A Gain-Invariant Baseline-Ratio Seizure Detector for Low-Cost Single-Channel EEG Wearables

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***Abstract*—Low-cost biopotential amplifiers and Wi-Fi microcontrollers now make a home EEG seizure alarm buildable by anyone, but the detection logic shipped with such devices is rarely tested against expert-annotated EEG. We present an open-source monitor built from an EXG BioAmp Pill front-end and an ESP32. It records one forehead channel (FP1–FP2) and alerts a phone over its own Wi-Fi hotspot without internet access. To test its detectors as they are actually written, we propose *code-faithful replay*: the CHB-MIT database (24 cases, 194 seizures, 220 h) is converted exactly to the device's montage, passed through a model of its analog front-end, and fed to the *unmodified* firmware logic and analyzer code under a simulated clock. Under event-based scoring with a chance-level baseline, the shipped analyzer detected only 5.7% of seizures. The firmware detector raised 18.29 false alarms per hour (FA/h), and random alarms at that rate would already detect 51.5% of seizures, against its own 75.3%. We trace both failures to short-memory adaptation, gain-dependent absolute thresholds and a feature vote dominated by uninformative spectral features. We then introduce *DEV-BR*, a baseline-ratio detector that reuses the signal the firmware already computes, needs only a 300-value running median and is invariant to amplifier gain. With parameters chosen on held-out patients, DEV-BR detected 47.4% of seizures (95% CI 30–71%) at 0.87 FA/h and 41.8% at 0.40 FA/h, outperforming the shipped analyzer in 21 of 24 patients and underperforming in none (p=6×10^{-5}). Hardware, firmware and the replay benchmark are open source.**

***Index Terms*—epilepsy, seizure detection, wearable EEG, consumer health technology, ESP32, edge computing, open hardware, reproducibility**

I. Introduction

Epilepsy affects about 50 million people worldwide [1]. Many seizures happen unwitnessed, especially at night. Unwitnessed convulsive seizures are a major risk factor for sudden unexpected death in epilepsy (SUDEP) [2], and patients' own diaries miss a large share of events [3]. The ILAE–IFCN guideline now recommends wearable seizure detectors, but only those validated on generalized tonic–clonic seizures [4]. Commercial wearables mostly sense movement and electrodermal activity [5], so they cannot see non-convulsive seizures, which are visible only in the EEG.

Open biopotential front-ends [6], [7] and microcontrollers such as the ESP32 [8] make a single-channel EEG alarm cheap and easy to build. The weak point of such designs is validation. Thresholds are tuned by hand, "pre-ictal" warning states are asserted without evidence, and algorithms are usually described or re-implemented rather than tested as the code that actually ships. A device is only as good as the code it runs, so this paper asks how well the device's code, exactly as shipped, detects seizures in annotated EEG, and what is the smallest change that makes it work. Our contributions are:

1) **An open, battery-powered single-channel EEG alarm** (front-end, ESP32 firmware, desktop analyzer, Android app) that alerts offline over its own access point (Sec. III).
2) **Code-faithful replay**, a validation method for consumer EEG devices. The shipped code runs unmodified against public EEG converted exactly to the device montage and passed through a model of its analog front-end, under a simulated clock. Results are scored by events with a chance-level baseline and patient-level confidence intervals (Sec. IV).
3) **A diagnosis** of why the shipped detectors fail, at the level of individual features and design choices (Sec. V).
4) **DEV-BR**, a gain-invariant baseline-ratio detector that fits in about 1 KB of microcontroller RAM. On held-out patients it detects about 7 times as many seizures as the shipped analyzer at a comparable false-alarm rate. We also report the effect of on-device personalization (Sec. VI).

II. Related Work

Automated seizure detection on scalp EEG goes back to Gotman's 1982 detector, which compared each epoch with its recent background [9]; Baumgartner and Koren review the field [10]. Line-length is a cheap and effective onset feature [11]. On CHB-MIT, patient-specific classifiers that use all 23 channels detect most seizures within seconds [12], and reduced montages such as behind-the-ear EEG keep useful but lower sensitivity [13]. Reporting practice has also matured. Event-based scoring with onset and offset tolerances, and false alarms normalized per hour or per day, are now recommended [14], [15]. Open amplifiers have been compared with clinical devices on signal quality [6], but to our knowledge the shipped detection code of a low-cost device has not been benchmarked unmodified against annotated seizures. We add a chance-level reference because a detector that alarms often enough appears sensitive by coincidence.

III. The Monitoring System

Hardware. A single differential channel between the left and right forehead (Fig. 1) is amplified by the BioAmp Pill [7], buffered and digitized at about 200 Hz. Forehead sites are hair-free, which suits gel-free consumer use, but they also pick up eye-movement artifacts. The device has no galvanic isolation, so it runs only from a battery bank. A GPIO line pulses for 500 ms on alarm (for a buzzer or Bluetooth trigger), and an RGB LED shows the current state. The bill of materials costs [TODO: BOM cost].

Firmware detector (shipped). For each 12-bit sample $x[n]$, the firmware tracks a baseline $b[n]{=}0.99\,b[n{-}1]{+}0.01\,x[n]$ and a deviation $d[n]{=}|x[n]-b[n]|$. While the state is NORMAL, d updates an exponentially weighted estimate of its mean μ and spread σ (forgetting factor 0.995 per sample). After a 90-s warm-up, the BROWNOUT (warning) and BLACKOUT (seizure) thresholds are reset every 30 s to $\mu{+}2.5\sigma$ and $\mu{+}4\sigma$, clamped to fixed ranges. Leaky counters require 20 or 15 supra-threshold samples before a transition. State, battery level and the waveform are served over a REST API on the ESP32's own access point.

Desktop analyzer (shipped). An optional Python analyzer reads the serial stream. Every 100 ms it computes line-length (LL), the delta/alpha power ratio (DAR) and the 30–45-Hz energy ratio (HFE) over a 2.56-s window. Each feature has adaptive thresholds at $\mu{+}2.5\sigma$ and $\mu{+}4\sigma$ with fixed floors. A state requires two of the three features to agree on 7 of 8 consecutive ticks. Epochs with LL above a ceiling are rejected as artifacts, and adaptation freezes during BLACKOUT.

Companion app. The Android app (Flutter) joins the device's hotspot and shows the current state, seizure duration, battery level and a live waveform. A simulation mode lets caregivers rehearse the alert flow.

IV. Code-Faithful Replay

Data and exact montage. We used the CHB-MIT Scalp EEG Database [16], [17]: 23 cases from 22 pediatric subjects aged 1.5–22 years, plus case chb24, sampled at 256 Hz with 16-bit resolution. Case chb21 is the subject of chb01, recorded 1.5 years later. We included every recording that contains a seizure plus at least one seizure-free recording per case, 198 seizures in 225 h. The device's FP1–FP2 derivation is not stored directly. We reconstructed it exactly as the signed sum of the bipolar chain FP1-F7, F7-T7, T7-FT9, FT9-FT10, FT10-T8, T8-F8, F8-FP2, or as FP1$-$FP2 for referential montages, wherever the electrodes exist. This gives 194 seizures in 220 h on FP1–FP2. FP1-F7 and FP2-F8 serve as secondary channels.

Front-end model. Each channel passes through a causal model of the analog pass-band: a first-order 0.5-Hz high-pass and a second-order 40-Hz Butterworth low-pass. It is then resampled to 200 Hz with a polyphase anti-aliasing filter, multiplied by the nominal gain of 1000 and quantized to 12 bits around mid-rail. Because the gain of an individual board is uncertain, we repeat the analysis at gains of 500 and 2000.

Running the shipped code. The published analyzer, `seizure_monitor.py`, is imported unmodified. Its own serial-reader loop is driven by a simulated serial port that emits the firmware's output format and a simulated clock that advances 5 ms per line, so warm-up, re-adaptation and confirmation delays behave exactly as in live use. Only plotting, logging and alerting are stubbed. The firmware's `loop()` is ported line by line, including its integer casts, and JIT-compiled. Its output forms the stream the analyzer consumes. Each recording starts from a fresh device state.

Scoring. Following SzCORE [15], an *alarm* is an entry into an alarm state. Alarms less than 90 s apart are merged. A seizure counts as detected if an alarm falls within [onset$-$30 s, offset$+$60 s]; every other alarm is a false alarm (FA). For each detector we report the sensitivity that a Poisson alarm process with the same rate λ would achieve, $1-e^{-\lambda(T+90\,\mathrm{s})}$ averaged over seizures of duration T. 95% confidence intervals come from a patient-level bootstrap (2000 resamples). Per-patient comparisons use the Wilcoxon signed-rank test.

V. Why the Shipped Detectors Fail

Outcomes. Table I gives the results. The analyzer's BLACKOUT alarm detected 11/194 seizures (5.7%) at 0.32 FA/h. Counting BROWNOUT warnings raised sensitivity to 21.1% but also raised FA/h to 2.82. The firmware detector's 75.3% sensitivity comes at 18.29 FA/h (438.9 per day), where random alarms would already reach 51.5%. Its spread estimate has a memory of about 1 s (forgetting factor 0.995 at 200 Hz), so a 4σ excursion lasting 75 ms is routine in ordinary EEG.

Feature evidence. Fig. 2 shows each feature's separation of ictal from inter-ictal ticks. Line-length (AUC 0.70; median per patient 0.79) and the firmware deviation d (AUC 0.70; per patient 0.83) carry seizure information. DAR (0.51) is at chance. HFE (0.40) points the wrong way. On a 0.5–40-Hz front-end the 30–45-Hz band is mostly filter roll-off, and what remains is likely dominated by forehead muscle activity, which is higher outside seizures in these recordings. The analyzer's 2-of-3 vote therefore requires agreement from features that carry no seizure information. Its LL floors (300 and 600 ADC units) are absolute values, so detection depends on amplifier gain. At gains of 500, 1000 and 2000, BLACKOUT sensitivity was 0.0%, 5.7% and 13.4% (Table II). Finally, the per-patient AUCs exceed the pooled ones, so each patient's background level differs and any fixed threshold suits only some patients.

VI. DEV-BR: A Baseline-Ratio Detector

The diagnosis suggests three requirements: normalize against the wearer's own slowly varying background rather than a 1-s window, require persistence on the order of seconds, and avoid absolute units. DEV-BR applies these to the deviation d, which the firmware already computes:

$$\delta(t) = \mathrm{mean}\{d[n] : n \in \text{second } t\}, \tag{1}$$

$$\beta(t) = \mathrm{median}\{\delta(\tau) : t{-}330 \le \tau < t{-}30\}, \tag{2}$$

$$\text{alarm} \iff \delta(\tau)/\beta(\tau) \ge r \;\; \forall\, \tau \in (t{-}D, t]. \tag{3}$$

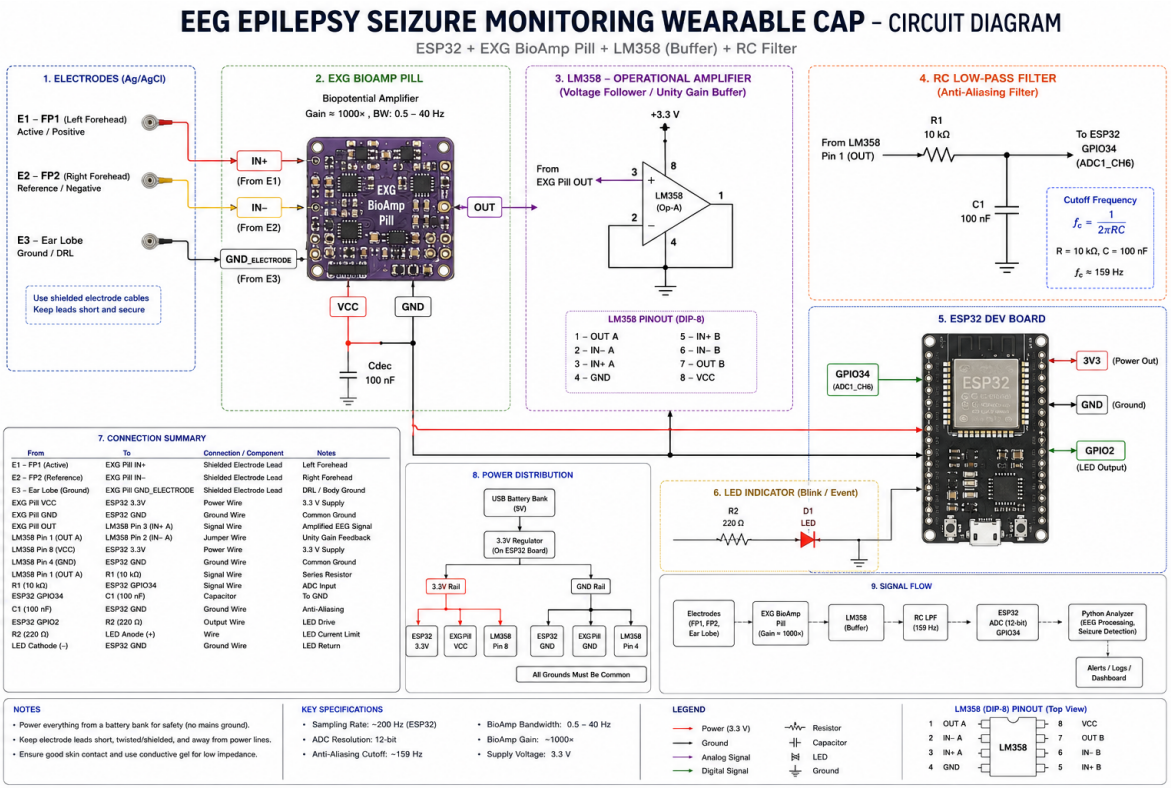


Fig. 1. Monitor hardware. Ag/AgCl electrodes at FP1 (+) and FP2 (–), with an earlobe ground, feed an EXG BioAmp Pill (gain ≈1000, pass-band 0.5–40 Hz). The output goes through an LM358 follower and an RC low-pass filter (10 kΩ/100 nF) into the ESP32's 12-bit ADC. The device runs from a USB battery bank.

TABLE I
EVENT-BASED RESULTS ON FP1–FP2 (194 SEIZURES, 220 H; FRONT-END 0.5–40 HZ, GAIN 1000). BRACKETS GIVE PATIENT-BOOTSTRAP 95% CIS. "CHANCE" IS THE SENSITIVITY RANDOM ALARMS AT THE SAME RATE WOULD ACHIEVE. RATIO DETECTORS (BR) ARE EVALUATED ON HELD-OUT PATIENTS ONLY.

Detector	Detected	Sensitivity [95% CI]	FA/h	Latency (s)	Chance
Analyzer, BLACKOUT (shipped)	11/194	5.7% [2–11]	0.32	26.3	1.3%
Analyzer, any alarm (shipped)	41/194	21.1% [14–30]	2.82	28.4	10.9%
Firmware, BLACKOUT (shipped)	146/194	75.3% [61–94]	18.29	14.5	51.5%
LL-BR, ≤1 FA/h	53/194	27.3% [15–44]	0.98	20.0	4.0%
FUSE-BR, ≤1 FA/h	72/194	37.1% [23–58]	0.81	23.5	3.3%
DEV-BR, ≤1 FA/h	92/194	47.4% [30–71]	0.87	18.0	3.5%
DEV-BR, ≤0.5 FA/h	81/194	41.8% [25–65]	0.40	23.0	1.7%
DEV-BR, ≤0.2 FA/h	59/194	30.4% [16–53]	0.18	25.0	0.7%

The 30-s lag keeps a seizure's onset out of its own baseline, and the median stays robust when a seizure fills part of the window. The ratio δ/β is unaffected, apart from quantization and clipping, by amplifier gain or any other constant scaling of the signal. The detector needs a 300-value buffer (1.2 KB at 32-bit) and one median per second. LL-BR applies the same rule to line-length, and FUSE-BR to the geometric mean of the two ratios. The parameters $r \in \{1.5, \ldots, 8\}$ and $D \in \{2, \ldots, 20\}$ s are chosen by patient-wise 2-fold cross-validation (odd versus even cases, with chb01 and chb21 together). On each training fold we pick the most sensitive setting that meets a target FA/h, and we report results only on the held-out fold.

Results. On held-out patients, DEV-BR detected 92/194 seizures (47.4%, CI 30–71%) at 0.87 FA/h with a median latency of 18.0 s. At stricter targets it detected 41.8% at 0.40 FA/h and 30.4% at 0.18 FA/h (Table I, Figs. 3 and 4). At a false-alarm rate comparable to the analyzer's BLACKOUT alarm, DEV-BR detected about 7 times as many seizures, and it did better in 21 of 24 patients and worse in 0 (p=6×10^{-5}). Against the analyzer's any-alarm output, DEV-BR had both higher sensitivity and a third of the false alarms (18 better, 2 worse, p=2×10^{-4}). It also beat LL-BR (15 better, 0 worse, p=7×10^{-4}). Fusing in line-length did not help. The ranking held on the secondary channels: DEV-BR detected 44.4% on FP1-F7 and 40.4% on FP2-F8 (Table II). Fig. 5 shows that the

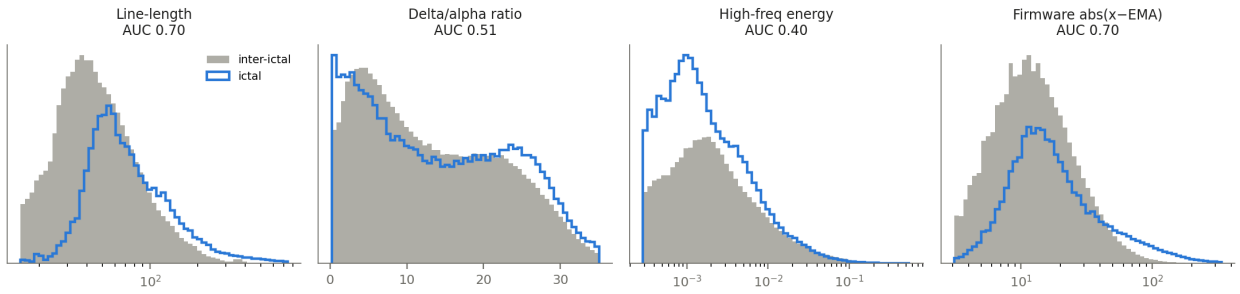


Fig. 2. Ictal (blue) versus inter-ictal (grey) distributions of the analyzer's three features and the firmware deviation d on FP1–FP2, with the pooled AUC.

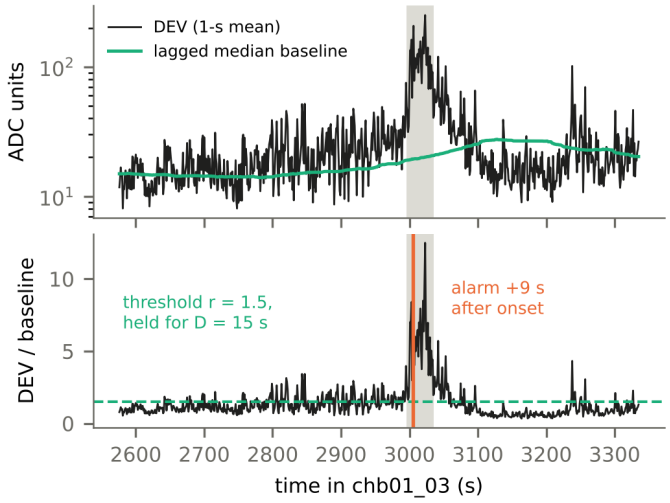


Fig. 3. DEV-BR on a seizure in chb01 (shaded). Top: the per-second deviation and its lagged median baseline. Bottom: their ratio, the held-out threshold, and the alarm.

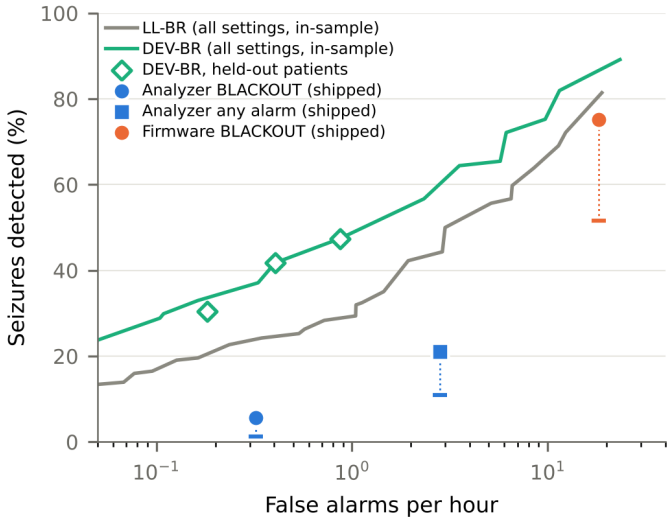


Fig. 4. Sensitivity versus false alarms per hour on FP1–FP2. Lines show the best in-sample trade-off over all ratio settings. Diamonds are DEV-BR on held-out patients. Markers are the shipped detectors, and the dash below each is the chance level at its alarm rate.

TABLE II
ROBUSTNESS. TOP: SENSITIVITY / FA/H ON FP1–FP2 AS THE FRONT-END GAIN VARIES (BLACKOUT FOR SHIPPED DETECTORS; LL-BR AT THE $\leq$0.5 FA/H TARGET AS A RATIO-BASED CONTROL). BOTTOM: SECONDARY CHANNELS AT GAIN 1000.

Gain	Analyzer	Firmware	LL-BR (CV)
×500	0.0% / 0.24	51.5% / 7.36	24.2% / 0.47
×1000	5.7% / 0.32	75.3% / 18.29	24.2% / 0.47
×2000	13.4% / 1.59	86.1% / 24.37	24.2% / 0.47

Channel	Analyzer (any)	Firmware	LL-BR	DEV-BR
FP1-F7	28.3% / 4.01	77.3% / 24.42	31.8% / 0.83	44.4% / 0.93
FP2-F8	29.3% / 3.83	82.8% / 24.79	30.8% / 1.04	40.4% / 0.67

gain is broad across patients and not driven by a few of them.

On-device personalization. We tested two consumer-feasible ways to adapt DEV-BR to the individual wearer, with every setting again chosen on other patients. (i) *Caregiver tap:* after one caregiver-marked seizure, the device sets its threshold from that seizure's peak ratio. On the remaining data (165 seizures, 184 h) this gave 52.1% at 1.15 FA/h versus 44.8% at 0.82 FA/h for population thresholds, a higher detection rate bought with more false alarms. (ii) *Wear-in:* the device uses its first two unlabeled recordings (71 h in total) to pick the lowest threshold that keeps its own alarm rate under a budget. This gave 51.9% at 0.72 FA/h versus 47.4% at 0.89 FA/h (156 seizures, 150 h), but no gain at the stricter target (30.1% versus 42.3%). Personalization therefore helps only modestly and inconsistently on this data. The large gain comes from using a baseline ratio rather than absolute thresholds.

VII. DISCUSSION AND LIMITATIONS

Code-faithful replay changed how the device's alarms should be described. The shipped on-device detector mostly fires at random. The analyzer's "pre-ictal" BROWNOUT state is a low-confidence warning with no demonstrated predictive value. The fix needs no new sensing: the firmware already computes the right signal and was missing only a baseline-ratio decision rule. We are moving DEV-BR into the firmware.

The absolute numbers also need stating plainly. DEV-BR detects about half of seizures at one false alarm per hour from a single forehead channel, below what clinical guidelines expect

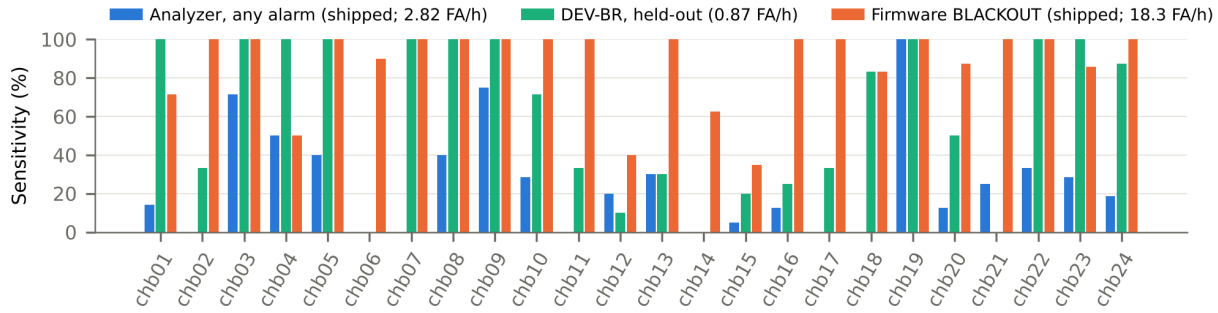


Fig. 5. Per-patient sensitivity on FP1–FP2 for the shipped analyzer (any alarm), DEV-BR at $\leq$1 FA/h (held-out) and the shipped firmware detector. The legend gives each detector's pooled false-alarm rate.

from a seizure alarm [4] and below multichannel patient-specific systems [12]. The monitor is a low-cost research and assistive prototype, not a medical device.

The study has several limitations. (i) CHB-MIT is a pediatric dataset recorded with clinical amplifiers, so adult, ambulatory and gel-free recordings will be noisier. (ii) The front-end is simulated, so electrode impedance, motion, 50-Hz mains interference and the weak anti-aliasing of the 159-Hz RC filter at 200-Hz sampling are not modelled. (iii) Recordings start cold, which penalizes seizures in the first minutes of a file. (iv) One frontal channel cannot see every seizure: seizures with posterior or deep onsets may not appear on it. (v) The Wi-Fi access point uses a fixed password, which must change before unattended use. The next steps are bench characterization (input-referred noise [TODO: μV_{rms}]; battery life [TODO: h]), on-device DEV-BR, and a supervised recording study under ethics approval.

VIII. Conclusion

This work shows that a single-channel EEG seizure alarm can be built from open, low-cost parts, and that its shipped detection code can be validated as it actually runs. Code-faithful replay on 24 CHB-MIT cases showed that the shipped detectors fail for identifiable reasons: short-memory adaptation, gain-dependent thresholds and a vote dominated by uninformative features. A gain-invariant baseline-ratio rule on a signal the firmware already computes detected 47.4% of seizures at 0.87 FA/h on held-out patients, a large and statistically significant improvement that fits in about 1 KB of microcontroller memory. The hardware, firmware, analyzer and replay benchmark are released so that other low-cost monitors can be validated the same way.

Data and Code Availability

CHB-MIT is available from PhysioNet [17]. All hardware files, firmware, the analyzer, the app and the replay benchmark (including a one-click Kaggle notebook) are at [18].

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